EXPERIMENTAL STUDY ON THE CARBON SEQUESTRATION POTENTIAL OF CLAYEY SOIL STABILIZED WITH FLY ASH

PROJECT PRELIMINARY REPORT

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF DEGREE OF

BACHELOR OF TECHNOLOGY

in

CIVIL ENGINEERING

of

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

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December 2026

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Certificate

This is to certify that the project report entitled **"EXPERIMENTAL STUDY ON THE CARBON SEQUESTRATION POTENTIAL OF CLAYEY SOIL STABILIZED WITH FLY ASH"** is a bona fide record of the work carried out by **Christofer C B (VAS23CE150)**, **Hariprasad T M (VAS23CE051)**, **Indraja Santhosh (VAS23CE021)**, and **Saniya E (VAS23CE036)** at Vidya Academy of Science & Technology, Thalakkottukara, Thrissur – 680 501, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Civil Engineering** of **APJ Abdul Kalam Technological University** during the academic year 2026–2027.

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December 2026

Institution Vision

To be a premier institution in engineering education and research, nurturing innovative professionals for sustainable societal development.

Institution Mission

M1: Provide quality engineering education that empowers students with strong technical competence and analytical problem solving abilities.

M2: Promote research, innovation, entrepreneurship and industry collaboration to develop sustainable technological solutions.

M3: Develop ethical and socially responsible professionals with leadership qualities, lifelong learning abilities and a commitment to societal well-being.

Department Vision

To excel in Civil Engineering education and research, moulding innovative and socially responsible professionals for sustainable development.

Department Mission

M1: Provide quality education in Civil Engineering that imparts technical knowledge and develops problem-solving skills.

M2: Promote research, innovation and entrepreneurship that contribute to sustainable infrastructure and societal development.

M3: Develop professionally competent Civil Engineers with ethical values, leadership qualities and commitment for societal well-being.

Department PEOs

PEO-1: Graduates will be able to apply knowledge of Civil Engineering principles, analytical skills, and modern engineering tools to provide innovative solutions for complex engineering problems.

PEO-2: Graduates will be able to excel in professional careers and higher studies by demonstrating technical competence, research aptitude, leadership qualities, and effective communication skills.

PEO-3: Graduates will be able to contribute to sustainable infrastructure development through innovation, entrepreneurship, and application of emerging technologies in Civil Engineering.

PEO-4: Graduates will be able to practice Civil Engineering with professional ethics, social responsibility, and a commitment to lifelong learning to address evolving societal and environmental challenges.

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PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/ develop systems/components/processes to

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PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

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PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

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PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

Acknowledgement

During the course of our Project work, several persons collaborated directly and indirectly with us. Without their support it would be impossible for us to do our work. That is why we wish to dedicate this section to recognize their support.

First and foremost I thank God Almighty for His providence and for being the guiding light throughout the Project.

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Finally, I would like to extend my sincere gratitude to friends who have always been helpful, in preparing and presenting the report and in the discussion following the presentation.

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Abstract

Carbon sequestration is a natural process in which carbon dioxide is captured from the atmosphere and stored in the soil as organic matter. It helps to mitigate climate change by reducing greenhouse gas concentrations, while improving soil fertility and ecosystem resilience. This project studies the carbon sequestration potential of clayey soil stabilized with alum (potassium aluminium sulphate). The review of literature shows that alum releases K^+ and Al^{3+} ions which compress the diffuse double layer of clay particles, bridge clay surfaces with organic matter and form macroaggregates that physically protect soil organic carbon. However, the combined action of the two ions, the optimum dosage, the durability of the bonds and the link between pore structure and carbon gas emission are not yet well understood.

The experimental programme starts with the collection of clayey soil and its characterization. The soil has a pH of 7.3, an electrical conductivity of 1.8 dS/m, a liquid limit of 60% and a plastic limit of 30%. Soil–alum specimens with 2, 4, 6, 8 and 10% alum were cast and tested for unconfined compressive strength after 1, 3 and 7 days of curing. The strength increased with curing period, and the maximum unconfined compressive strength of 195 kPa was obtained at 6% alum after 7 days of curing. The next stage of the work is the exposure of the specimens to carbonation and the measurement of carbon uptake, followed by comparison of the carbon sequestration capacity of the soil before and after treatment.

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Chapter 1

INTRODUCTION

This chapter introduces the project on the carbon sequestration potential of clayey soil stabilized with alum. It explains why soil carbon sequestration is important, the Sustainable Development Goals addressed by the work, the objectives and scope of the project, and the methodology adopted.

1.1 Introduction

Carbon sequestration is a critical natural process in which carbon dioxide is captured from the atmosphere and stored in the soil as organic matter. This process plays a vital role in mitigating climate change by reducing greenhouse gas concentrations, while simultaneously improving soil fertility and ecosystem resilience.

1.1.1 Importance of Carbon Sequestration

The importance of soil carbon sequestration can be summarized as follows.

- Atmospheric CO_2 has exceeded 420 ppm [14].
- Global warming is increasing rapidly.
- Soil stores almost three times more carbon than the atmosphere [4].
- Carbon sequestration improves both environmental quality and soil fertility.

1.1.2 Sustainable Development Goals Mapped

The project contributes to the following Sustainable Development Goals (SDGs), as shown in Table 1.1.

Table 1.1: SDGs mapped to the project

SDG	Contribution of the project
SDG 2 – Zero Hunger	Healthy soils improve crop productivity
SDG 6 – Clean Water and Sanitation	Improves infiltration and reduces runoff
SDG 13 – Climate Action	Reduces atmospheric CO_2
SDG 15 – Life on Land	Improves biodiversity and soil health

1.2 Objectives of the Work

The objectives of the project are:

1. To find the chemical index and engineering properties of the collected soil sample.
2. To characterise the properties of the procured alum.
3. To find the optimum dosage of alum for maximum soil stabilization.
4. To compare the carbon sequestration capacity of the clay soil before and after the treatment.

1.3 Scope of the Project

This project focuses on studying the carbon sequestration potential of clayey soil stabilized with alum powder. It examines how different amounts of alum affect the properties and strength of clayey soil and identifies the optimum alum content. The study also evaluates the ability of the treated soil to capture and store carbon, while promoting the reuse of alum as a sustainable material for soil stabilization and environmental protection.

1.4 Motivation for this work

The atmospheric CO_2 concentration is rising and soil is a large natural store of carbon, so any method that increases the carbon retained in soil is useful for climate change

mitigation. Chemical stabilization of clayey soil with alum is known to improve the structure of the soil, but its effect on carbon retention has not been clearly established. The research gaps identified from the literature (Section 2.14) motivate an experimental study of the strength and carbon sequestration behaviour of alum-stabilized clayey soil. The work also promotes the reuse of alum as a sustainable material for soil stabilization.

1.5 Methodologies Adopted

The work is carried out in two phases. In Phase 1, raw clayey soil is collected, characterized physically, geotechnically and chemically, the mix proportions are designed and the carbonation chamber (exposure environment) is set up. In Phase 2, soil–alum specimens are cast and compacted, exposed to carbonation, tested for strength, and the carbon uptake and sequestration potential are measured and compared. The methodology is described in detail in Chapter 3.

1.6 Outline of the Report

This report contains 6 chapters. Chapter 1 gives the introduction to the project work and describes the objectives and scope of the work. The literature survey, the summary of the literature and the research gaps identified are described in Chapter 2. Chapter 3 explains the methodology, the materials and the tests used in the project. Chapter 4 presents the experimental investigations carried out and the results obtained with discussion. Chapter 5 describes the project status completed in S7, and Chapter 6 describes the work to be completed in S8 along with the project timeline, expected outcomes and future scope.

1.7 Summary

This chapter introduced carbon sequestration in soil, its importance, the SDGs addressed, the objectives, the scope and the methodology of the project. The next chapter reviews the literature on carbon sequestration in clayey soil stabilized with alum.

Chapter 2

LITERATURE REVIEW

This chapter presents twelve research papers reviewed for the project. For each paper the journal, title, method used and the main research finding are given. A summary of the review and the research gaps identified from the literature are given at the end of the chapter.

2.1 Thomas et al. (2026) [12]

Journal: Sustainability

Title: Sustainable Chemical Amendments for Soil Health and Carbon Retention

Method: Volumetric swelling dynamics testing and soil gas diffusion measurements

Finding: Inorganic alum-based salts reduce swelling and moisture dispersion, creating a dense chemical matrix that restricts carbon gas diffusion from the soil.

2.2 Meena (2026) [10]

Journal: International Journal of Plant & Soil Science

Title: Role of Soil Organic Carbon Sequestration in Climate Change Mitigation and Sustainable Agriculture: A Comprehensive Review

Method: Soil organic matter protection mechanism evaluation

Finding: Alum salts promote clay–polyvalent cation–organic matter (C-P-OM) complexation, enhancing the long-term carbon storage capacity of fine-textured soils.

2.3 Nazir et al. (2026) [11]

Journal: Journal of Global Innovations in Agricultural Sciences

Title: Soil Carbon Sequestration as a Climate Change Mitigation Strategy: A Review

Method: Cation exchange capacity (CEC) and aggregate turnover rate analysis

Finding: Monovalent K^+ and trivalent Al^{3+} ions act as structural anchors with clay particles, significantly lowering the decomposition rate of sequestered carbon.

2.4 Karakan and Yilmaz (2026) [7]

Journal: Applied Sciences

Title: Performance of Chemical-Stabilized Clayey Soil Subjected to Environmental Durability Cycles

Method: Wetting–drying and freeze–thaw durability cycling and Atterberg limits testing

Finding: K^+ and Al^{3+} ions compress the diffuse double layer of clay platelets, inducing immediate flocculation and preventing carbon loss during weathering cycles.

2.5 Wang et al. (2025) [13]

Journal: Agronomy

Title: Aluminum Sulfate and Straw Enhance Carbon Sequestration in Saline–Alkali Soils

Method: Laboratory soil incubation and wet-sieving aggregate fractionation

Finding: Hydrated Al^{3+} hydrolyzes into monomeric and polymeric species, promoting macroaggregate formation and locking soil organic carbon and heavy fraction organic carbon (HFOC).

2.6 Das et al. (2025) [2]

Journal: Geoderma / Authorea

Title: Soil Carbon Sequestration: A Mechanistic Perspective on Limitations and Future Possibilities

Method: Spectroscopic analysis of cation bridging (Al^{3+}–clay–SOC bonds)

Finding: K^{+} and Al^{3+} ions compress the diffuse double layer of clay platelets, inducing immediate flocculation and preventing carbon loss during weathering cycles.

2.7 Ghimirey et al. (2025) [3]

Journal: Discover Soil

Title: Understanding Soil Carbon Sequestration: Mechanistic Insights, Management Approaches, and Future Challenges

Method: Comprehensive systematic review and organo-mineral stability modeling

Finding: Alum treatments provide dissolved Al^{3+} cations that bind fine clay particles, forming organo-mineral complexes that physically isolate carbon from microbial oxidation.

2.8 Jang et al. (2025) [6]

Journal: Canadian Geotechnical Journal

Title: Role of Calcined-Clay-Based Geopolymers for Effective Stabilization of Expansive Soils

Method: Unconfined compressive strength (UCS) testing and SEM/EDX microstructural imaging

Finding: Multi-cation systems with components of potash alum induce clay flocculation, compressing pores and physically encapsulating carbon.

2.9 Antunes et al. (2025) [1]

Journal: Revista Brasileira de Ciências Ambientais

Title: Soil Carbon Storage and Retention: Sustainable Application in Environmental Engineering

Method: Geotechnical laboratory testing and micro-porosity analysis by mercury intrusion porosimetry

Finding: Integrates clay stabilization with environmental carbon retention, showing that alum chemical additives increase the micro-porous shielding of organic carbon.

2.10 Kumaragamage et al. (2024) [9]

Journal: Catena

Title: Residual Benefits of Alum, Gypsum, and Magnesium Sulfate Amendments in Reducing Phosphorus Losses to Snowmelt Runoff

Method: Simulated snowmelt runoff and microstructural surface complexation analysis

Finding: Alum introduces polyvalent cations that induce surface complexation on clay colloids, which suppresses soil dispersion and stabilizes organo-mineral structures.

2.11 Knight et al. (2024) [8]

Journal: Journal of Environmental Quality

Title: Phosphorus Fractions and Speciation in an Alkaline, Manured Soil Amended with Alum, Gypsum, and Epsom Salt

Method: X-ray absorption near-edge structure (XANES) and chemical fraction sequential extraction

Finding: Integrates clay stabilization with environmental carbon retention, showing that alum chemical additives increase the micro-porous shielding of organic carbon.

2.12 Indraratne et al. (2024) [5]

Journal: Canadian Journal of Soil Science

Title: Loss of Potentially Toxic Elements to Snowmelt Runoff from Soils Amended with Alum, Gypsum, and Epsom Salt

Method: Soil column leaching tests and diffuse double layer (DDL) microstructural profiling

Finding: Direct alum application compresses the DDL of clay minerals, acting as an abiotic binder that prevents fine particle detachment and traps organic carbon.

2.13 Summary of Literature Review

The main points drawn from the reviewed literature are as follows.

- **Diffuse Double Layer (DDL) Compression:** Alum releases high-valence ions that neutralize clay surface charges, causing rapid particle aggregation and structural stability.
- **Cation Bridging:** Polyvalent Al^{3+} acts as a strong ionic bridge linking clay mineral surfaces directly to organic matter.
- **Carbon Encapsulation:** Chemical stabilization creates macroaggregates that physically lock in soil organic carbon (SOC), protecting it from microbial breakdown.
- **Reduced Gas Diffusion:** Microstructural compaction decreases pore space, lowering the rate at which carbon dioxide (CO_2) escapes from the soil.

2.14 Research Gaps Identified from the Literature

The following research gaps were identified.

- **Dual-Ion Synergy:** How potassium (K^+) and aluminium (Al^{3+}) act together during alum breakdown to affect long-term carbon storage remains unclear.

- **Optimal Dosage Threshold:** Existing studies lack a clear dosage limit that maximizes carbon entrapment without making the clay soil too brittle.
- **Environmental Durability:** The durability of Al^{3+}–clay–SOC bonds under repeated wetting–drying and freeze–thaw cycles is not fully understood.
- **Microstructure vs. Gas Kinetics:** Direct links connecting pore network changes (measured through porosimetry) to subsoil carbon gas emissions are insufficiently explored.

2.15 Summary

This chapter reviewed twelve papers on the role of alum and its cations in clay stabilization and soil carbon retention. The review showed that alum improves soil structure and protects organic carbon, and it identified gaps related to dual-ion action, dosage, durability and gas kinetics. The next chapter describes the methodology adopted to address these gaps.

Chapter 3

METHODOLOGIES FOR THE PROJECT

This chapter describes the methodology of the project, the materials and equipment required and the tests to be carried out on the soil.

3.1 Methodology

The methodology is divided into two phases with five steps in each phase, as shown in Figure 3.1.

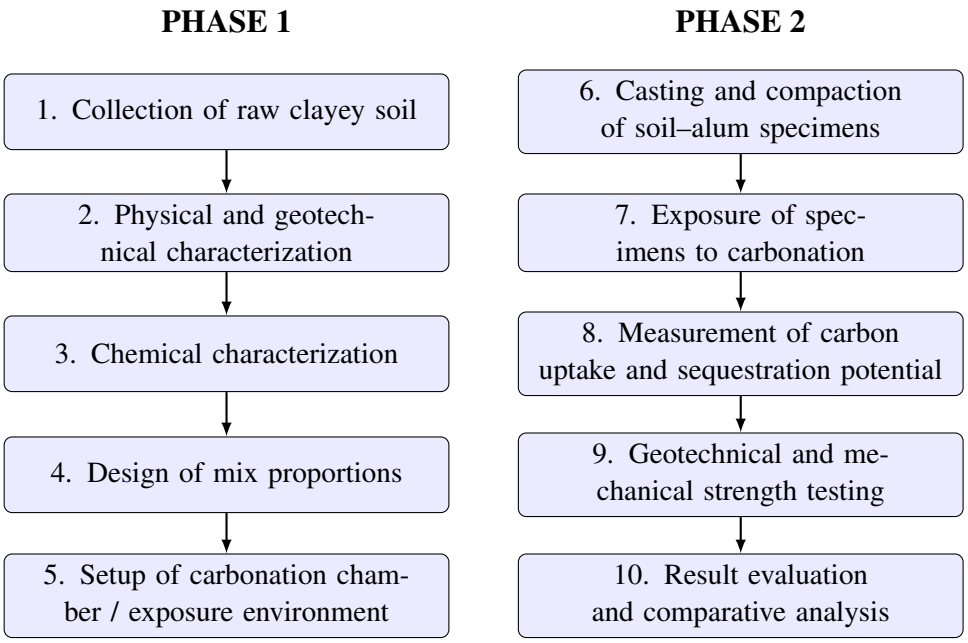


Figure 3.1: Methodology of the project

3.1.1 Phase 1

1. Collection of raw clayey soil.
2. Physical and geotechnical characterization of the soil.
3. Chemical characterization of the soil.
4. Design of the mix proportions.
5. Setup of the carbonation chamber / exposure environment.

3.1.2 Phase 2

6. Casting and compaction of the soil–alum specimens.
7. Exposure of the specimens to carbonation.
8. Measurement of carbon uptake and sequestration potential.
9. Geotechnical and mechanical strength testing.
10. Result evaluation and comparative analysis.

3.2 Materials Required

- Clayey soil
- Alum (potassium aluminium sulphate)
- Distilled water
- CO_2 source / controlled exposure system

3.3 Equipment Required

- Weighing balance
- Oven
- Sieve set

- Mixing trays
- Containers
- Measuring cylinders

3.4 Soil Testing

The following tests are carried out on the soil.

1. Atterberg limits
2. Sieve analysis
3. Natural water content
4. Specific gravity
5. pH test
6. Electrical conductivity test
7. Unconfined compression (UCC) test
8. Free swell test

3.5 Summary

This chapter explained the two-phase methodology, the materials, the equipment and the tests used in the project. The next chapter presents the experimental work carried out and the results obtained.

Chapter 4

EXPERIMENTAL INVESTIGATIONS AND RESULTS

This chapter presents the laboratory tests performed on the clayey soil and on the soil–alum specimens, and the results obtained with discussion.

4.1 Tests Performed

4.1.1 Soil Collection and Characterization Tests

Clayey soil was collected from the field and the characterization tests shown in Figure 4.1 were carried out in the laboratory.

4.1.2 Unconfined Compression (UCC) Test

Soil–alum specimens were prepared and tested in unconfined compression. The preparation and testing of the specimens are shown in Figure 4.2.

4.2 Results and Discussion

4.2.1 Properties of the Clayey Soil

The properties obtained for the clayey soil are given in Table 4.1.

The plasticity index of the soil is obtained from the liquid limit (LL) and the plastic limit (PL) as



(a) Soil collection



(b) Sieve size analysis



(c) Atterberg limits



(d) Specific gravity



(e) Free swell test



(f) pH test and electrical conductivity

Figure 4.1: Soil collection and characterization tests

Table 4.1: Properties of the clayey soil

Property	Value
pH	7.3
Electrical conductivity	1.8 dS/m
Liquid limit	60%
Plastic limit	30%

$$I_P = LL - PL = 60 - 30 = 30\% \tag{4.1}$$

4.2.2 UCC Test Results

The UCC tests were carried out on clayey soil stabilized with 2, 4, 6, 8 and 10% alum after curing periods of 1, 3 and 7 days. The axial stress values obtained at different axial strains are given in Tables 4.2, 4.3 and 4.4.

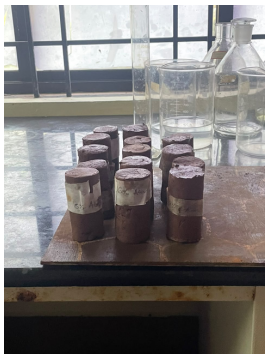
The unconfined compressive strength (q_u) is the peak axial stress of each specimen. The values obtained for the different alum contents and curing periods are summarized in Table 4.5 and the stress–strain curves and the variation of q_u with alum content are



(a) Specimen preparation



(b) Mixing of soil and alum



(c) Specimens



(d) Specimens in desiccator



(e) UCC testing

Figure 4.2: Preparation and testing of soil–alum specimens (UCC test)

shown in Figure 4.3.

4.2.3 Discussion

- At every curing period the unconfined compressive strength increased as the alum content was raised from 2% to 6% and then decreased at 8% and 10% alum. The highest strength was obtained at 6% alum, which is therefore the optimum alum content among the dosages tested.
- For every alum content the strength increased with the curing period. For 6% alum, q_u increased from 125 kPa after 1 day to 160 kPa after 3 days and 195 kPa after 7 days.
- The maximum unconfined compressive strength of 195 kPa was obtained for 6% alum after 7 days of curing.

Table 4.2: UCC test: axial stress (kPa) of clay soil stabilized with alum, 1 day of curing

Axial strain (%)	2% Alum	4% Alum	6% Alum	8% Alum	10% Alum
0	0	0	0	0	0
0.5	9.1	10.4	10.4	9	7
1	27.7	32	32.7	28.4	22.5
1.5	47.5	56.2	58.4	51.5	41.4
2	63.5	76.8	81.6	73.1	59.7
2.5	74.1	91.7	99.3	90.5	75.1
3	85	100.9	111.1	102.9	86.7
3.5	81.3	110	125	110.9	94.5
4	80.8	104.9	119.5	120	105
5	79.8	103.6	118	113.7	99.8
6	78.7	102.2	116.6	112.2	98.5
8	76.7	99.6	113.6	109.4	96
10	74.7	97	110.6	106.5	93.5

Table 4.3: UCC test: axial stress (kPa) of clay soil stabilized with alum, 3 days of curing

Axial strain (%)	2% Alum	4% Alum	6% Alum	8% Alum	10% Alum
0	0	0	0	0	0
0.5	11.2	13.2	13.4	11.6	9.1
1	34.2	40.8	41.9	36.7	29
1.5	58.7	71.5	74.8	66.5	53.2
2	78.4	97.8	104.4	94.5	76.8
2.5	91.5	116.7	127	117	96.6
3	105	128.4	142.2	133	111.5
3.5	100.4	140	160	143.2	121.5
4	99.8	133.5	153	155	135
5	98.5	131.8	151.1	146.8	128.3
6	97.3	130.1	149.2	145	126.7
8	94.8	126.8	145.3	141.3	123.4
10	92.2	123.4	141.5	137.5	120.2

4.3 Summary

This chapter presented the characterization of the clayey soil (pH 7.3, electrical conductivity 1.8 dS/m, liquid limit 60%, plastic limit 30%) and the UCC test results of soil stabilized with 2 to 10% alum. The strength was maximum at 6% alum and increased with curing time. The next chapter summarizes the status of the project at the end of S7.

Table 4.4: UCC test: axial stress (kPa) of clay soil stabilized with alum, 7 days of curing

Axial strain (%)	2% Alum	4% Alum	6% Alum	8% Alum	10% Alum
0	0	0	0	0	0
0.5	13.4	16	16.3	13.8	10.7
1	40.7	49.5	51.1	43.8	34.4
1.5	69.8	86.8	91.1	79.3	63.1
2	93.3	118.8	127.3	112.7	91
2.5	108.9	141.7	154.8	139.6	114.5
3	125	155.9	173.3	158.7	132.1
3.5	119.5	170	195	170.9	144
4	118.8	162.1	186.5	185	160
5	117.3	160	184.2	175.3	152.1
6	115.8	158	181.8	173	150.1
8	112.8	153.9	177.1	168.6	146.3
10	109.8	149.8	172.5	164.2	142.5

Table 4.5: Unconfined compressive strength (q_u) of clay soil stabilized with alum

Alum content (%)	1 day q_u (kPa)	3 days q_u (kPa)	7 days q_u (kPa)
2	85	105	125
4	110	140	170
6	125	160	195
8	120	155	185
10	105	135	160

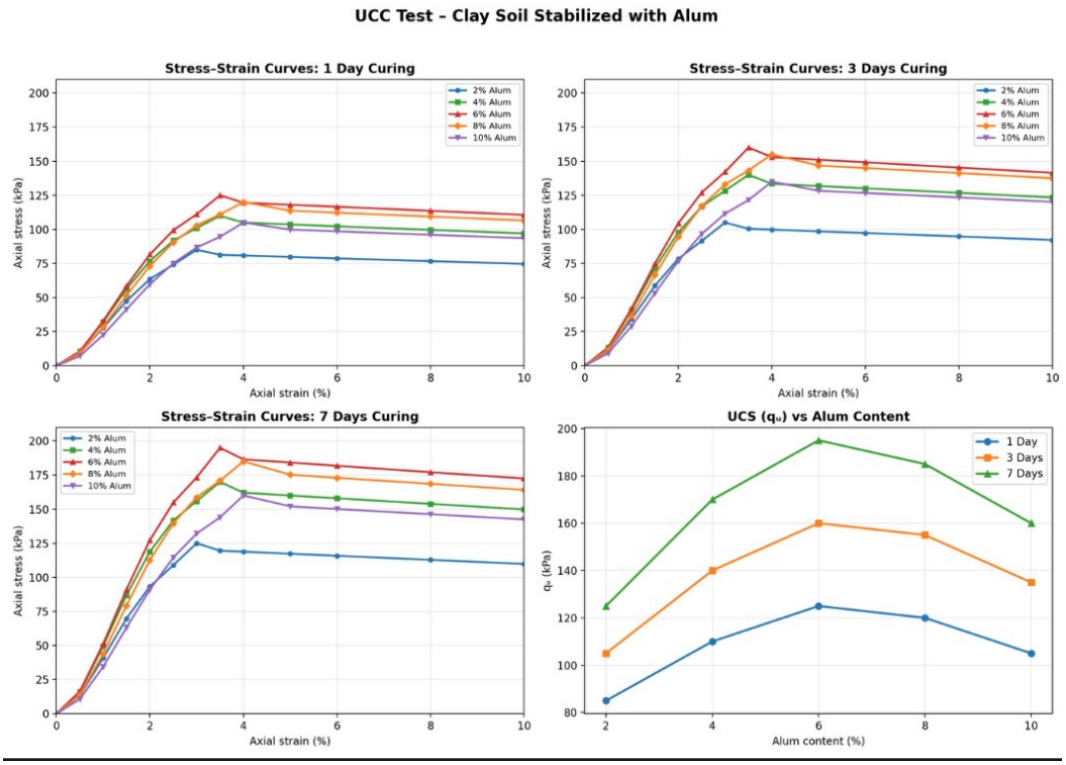


Figure 4.3: UCC test: stress–strain curves for 1, 3 and 7 days of curing and variation of q_u with alum content

Chapter 5

PROJECT STATUS COMPLETED IN S7

The work completed in the seventh semester is listed below.

- Literature survey of twelve research papers and identification of the research gaps (Chapter 2).
- Definition of the objectives and scope of the project.
- Collection of raw clayey soil from the field.
- Physical, geotechnical and chemical characterization of the soil: sieve analysis, Atterberg limits, specific gravity, free swell test, pH and electrical conductivity.
- Design of mix proportions with 2, 4, 6, 8 and 10% alum, and casting and compaction of the soil–alum specimens.
- Unconfined compression tests after 1, 3 and 7 days of curing, and identification of 6% alum as the optimum content among the dosages tested (Chapter 4).

Chapter 6

PROJECT STATUS TO BE COMPLETED IN S8

6.1 Work Planned

The remaining work consists of the carbonation stage and the analysis of the results.

- Setup of the carbonation chamber / exposure environment.
- Exposure of the specimens to carbonation.
- Measurement of carbon uptake and sequestration potential.
- Comparison of the carbon sequestration capacity of the clay soil before and after the treatment.
- Result evaluation, comparative analysis and final report preparation.

6.2 Project Timeline

The planned timeline of the project, in seven research phases, is given in Table 6.1.

6.3 Expected Outcomes

- Improvement in the engineering properties of clayey soil with suitable alum addition.

Table 6.1: Project timeline

Sl. No.	Research phase	Period	Duration (months, approx.)
1	Literature review	Aug–Sep	1.5
2	Material characterization	Mid Sep–Oct	1.25
3	Mix design and sample preparation	Oct–Mid Nov	1.25
4	Carbon sequestration and curing experiments	Nov–Dec	2
5	Post-curing geotechnical testing	Mid Dec–Jan	1.5
6	Microstructural analysis	Jan–Mid Feb	1.25
7	Final presentation	Mid Feb–Mar	1

- Increase in soil strength at an optimum alum content.
- Demonstration of measurable carbon sequestration / CO_2 uptake potential.
- Effective utilization of alum as an industrial waste material.
- Development of a more sustainable approach to clayey soil stabilization.

6.4 Future Scope

- Utilization of different types of alum from different sources.
- Long-term monitoring of carbon storage.
- Economic feasibility and potential carbon-credit applications.
- Development of sustainable soil stabilization strategies combining waste utilization and climate mitigation.

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